Comparative Unlabelled Anomaly Detection in Daily Climate Time Series

Abstract

We compare a seasonal burst detector, three discord/profile methods, Isolation Forest, and three deep reconstruction models on daily temperature, total precipitation, and pressure. All models are fitted on 1970–1999 and evaluated on 2000–2023. Score detectors are compared at a detector-specific empirical 99th percentile without rescaling their scores. Raw exceedances are used for pairwise Jaccard agreement; temporally declustered peaks are used to define and interpret events. The results show strong dependence on both variable and anomaly definition. Total precipitation produces the strongest multi-method agreement, while the z-normalized matrix profile is largely complementary to amplitude and reconstruction methods. Very small reconstruction loss, most notably for DLinear, does not imply stronger event validity. In the absence of labels, the conclusions concern detector behaviour and reproducibility rather than physical correctness.

1 Objectives

The study addresses three questions: (i) how detector scores cluster into events after a common tail-selection rule; (ii) which raw anomalies and events are shared across detector families; and (iii) how those patterns differ among temperature, total precipitation, and pressure. The analysis is deliberately unlabelled. Agreement is therefore interpreted as robustness to method choice, not as proof that an event is a true physical anomaly.

2 Experimental protocol

2.1 Data preparation and split

The inputs are daily arrays named `temperature_time_series.npy`, `preci_time_series.npy`, and `pressure_time_series.npy`. A Gregorian date index is constructed, 29 February is removed when present, and only complete years in the common period 1 January 1970–31 December 2023 are kept. This produces 54 years of 365 observations. The first 30 years (1970–1999; 10,950 days) form the reference period used for fitting, and the remaining 24 years (2000–2023; 8,760 days) form the evaluation period. Reference-period statistics alone are used for standardization and reference-neighbour searches.

This split prevents direct evaluation-to-reference leakage. It does not remove the effect of climatic trends or other distribution shifts, which are part of what the detectors may score as abnormal. The source metadata and physical units are not encoded in the current numerical outputs, so physical-unit reconstruction errors are reported within each variable but not compared across variables. For readability, physical-data figures convert temperature from K to °C, mean sea-level pressure from Pa to hPa, and total precipitation from m to mm. These display conversions do not alter detector scores or event selection.

2.2 Detectors and fitting

All score outputs retain their native scale; in particular, they are not rescaled to $[0, 1]$. Larger scores always denote greater abnormality.

Seasonal bursts.
Evaluation aggregates of length 1–15 days are compared with reference aggregates at the same seasonal position. Absolute standardized departures of at least 3 are retained and overlapping same-sign candidates are merged. Bursts already return intervals and do not undergo percentile thresholding.

Subsequence and ensemble detectors.
The classical subsequence length is $m = 15$ days. IMP uses an interval length of 30 days. MP uses z-normalized subsequences and is consequently sensitive to shape discord; AAMP uses unnormalized distance and retains level and amplitude information. The nearest reference subsequence is stored for each profile score. These methods build on the matrix-profile framework [2]. Isolation Forest [1] uses 300 trees and seed 0 on overlapping 15-day windows; its output sign is reversed so that large values are anomalous.

Deep reconstruction detectors.
AutoEncoder [3], DLinear [4], and TimesNet [5] receive standardized sequences of length 96. Their overlapping reconstructions are averaged into a single time series, and the daily anomaly score is the corresponding overlapping mean squared error. AutoEncoder hidden-width ratios are 0.5 and 0.25. DLinear uses a moving-average ratio of 0.1 (an odd kernel of 9 for length 96). TimesNet uses $d_{\text{model}} = 16$, $d_{\text{ff}} = 32$, two layers, top-$k = 3$, six convolution kernels, and dropout 0.1. Each model is trained for 50 epochs using Adam, a learning rate of 10^{-2}, batch size 1024, seed 0, and the last 10% of the reference-period sequences as a temporal validation set. No early stopping or validation-based checkpoint selection is used; reported losses are from the final model.

2.3 Tail selection and declustering

For detector d, let $s_{d,t}$ denote its evaluation score and let $q_d = Q_{0.99}(s_d)$ be its empirical 99th percentile. A day is a raw anomaly when

$$I_{d,t} = \mathbf{1}\{s_{d,t} > q_d\}. \tag{1}$$

Thus the rule selects scores *strictly above* the percentile, rather than sorting and forcing an exact top-k quota. Ties at the threshold can therefore make the number differ from one percent. In this run all continuous score series yield 88 raw exceedances, approximately 1% of 8,760 evaluation days.

Raw exceedances separated by at most $m = 15$ days belong to the same run. Each run is represented by its maximum-score day. If N_d is the raw count and K_d the declustered count, we report

$$\theta_d = K_d/N_d. \tag{2}$$

Small θ_d denotes persistent or recurrent threshold exceedances; large θ_d denotes more isolated exceedances. No seasonal threshold is fitted and no separate set of seasonal peaks is created.

Seasonality is characterized on the complete score series with an STL decomposition [6] of period 365, $s_t = T_t + S_t + R_t$. The reported seasonal strength is

$$F_S = \max\left(0, \min\left(1, 1 - \frac{\operatorname{Var}(R_t)}{\operatorname{Var}(S_t + R_t)}\right)\right). \tag{3}$$

It measures annual organization in the score process; it does not alter the threshold or the selected anomalies.

2.4 Agreement and consensus

Pairwise Jaccard agreement is computed on the *non-declustered* raw flags,

$$J_{ij} = \frac{|E_i \cap E_j|}{|E_i \cup E_j|}, \qquad E_i = \{t : I_{i,t} = 1\}. \tag{4}$$

When two score arrays have different lengths, only their shared dated support is used; missing trailing dates are not counted as normal observations.

For the event timeline, each declustered representative marks the following 15 days. A selected day for method i is a consensus day if at least one other method is also active, and the method-level consensus rate is the fraction of its selected days that are consensus days. The separate consensus-event table uses a symmetric ± 15-day neighbourhood and requires at least three active score detectors. Its `n_detectors` field is the simultaneous count on the event peak; the detector list records any method participating anywhere in the merged interval and can therefore be longer.

2.5 Interpretation views

Interpretation is performed on declustered representatives. Point detectors mark the cluster peak and any other raw exceedance from that cluster that falls inside the displayed window. Deep-model companion points outside the window are not shown. For IMP, MP, and AAMP, an anomaly is a subsequence: the complete 15-day discord beginning at the cluster peak is displayed in three panels—a folded evaluation view, the nearest-neighbour folded reference view, and their direct overlay. For MP, the two folded views retain the original data scale and share vertical limits; only the direct discord–neighbour comparison is separately z-normalized. IMP and AAMP retain native scale in all three panels. Bursts remain intervals. At most five events per method and variable are interpreted.

3 Results

3.1 Peak extraction and score seasonality

Table 1 reports the complete peak diagnostics. Because the raw tail budget is the same, differences in event counts are entirely due to the temporal arrangement of each detector's large scores.

Table 1: Raw exceedances, declustered events, event ratio θ, and STL seasonal strength F_S.

Series	Method	Raw	Events	θ	F_S
Temperature	IMP	88	17	.193	.264
	MP	88	36	.409	.262
	AAMP	88	17	.193	.260
	Isolation Forest	88	13	.148	.404
	AutoEncoder	88	67	.761	.080
	DLinear	88	5	.057	.803
	TimesNet	88	15	.170	.000
Total precipitation	IMP	88	7	.080	.189
	MP	88	28	.318	.313
	AAMP	88	9	.102	.160
	Isolation Forest	88	12	.136	.177
	AutoEncoder	88	74	.841	.339
	DLinear	88	62	.705	.368
	TimesNet	88	62	.705	.303
Pressure	IMP	88	18	.205	.517
	MP	88	32	.364	.262
	AAMP	88	14	.159	.466
	Isolation Forest	88	19	.216	.531
	AutoEncoder	88	44	.500	.092
	DLinear	88	33	.375	.199
	TimesNet	88	52	.591	.529

Clustering depends strongly on both method and variable. Temperature DLinear has only

five representatives (0.06), so its top errors occur in a few long or recurrent runs; its score is also strongly annual ($F_S = 0.80$). In contrast, temperature AutoEncoder produces 67 relatively isolated events and weak score seasonality. Total precipitation shows a detector-family split: IMP, AAMP, and Isolation Forest compress 88 days into only 7–12 events, whereas the deep models retain 62–74 events. Pressure is intermediate. Its IMP, AAMP, Isolation Forest, and TimesNet scores all have substantial annual organization ($F_S = .466$–$.531$), which cautions against interpreting a global-tail event list as seasonally uniform.

3.2 Agreement

Table 2 gives the number of events and the percentage of each method's selected timeline days overlapping at least one other method.

Table 2: Declustered event count and method-level consensus rate C (percent).

	Temperature		Total precipitation		Pressure	
Method	Events	C	Events	C	Events	C
Burst	58	45.4	252	55.4	57	46.7
IMP	17	92.5	7	100.0	18	81.5
MP	36	18.5	28	43.6	32	11.9
AAMP	17	85.9	9	99.3	14	93.3
Isolation Forest	13	73.3	12	98.9	19	76.8
AutoEncoder	67	22.9	74	82.5	44	70.5
DLinear	5	57.3	62	92.0	33	49.5
TimesNet	15	53.3	62	77.2	52	55.3

The non-declustered Jaccard matrices make the family structure clearer. For total precipitation, IMP–AAMP has $J = .743$, AAMP–Isolation Forest $J = .645$, and IMP–Isolation Forest $J = .530$. The deep family forms a second block: AutoEncoder–DLinear $J = .544$, AutoEncoder–TimesNet $J = .375$, and DLinear–TimesNet $J = .354$. Cross-family overlaps are much smaller. For temperature the largest overlap is IMP–AAMP ($J = .257$), followed by IMP–Isolation Forest ($J = .222$); for pressure it is again IMP–AAMP ($J = .419$). MP has zero raw overlap with every other score detector for pressure and nearly zero overlap for temperature. This is consistent with MP detecting z-normalized shape discord rather than level or amplitude extremes.

Total precipitation has the strongest event-level reproducibility: the mean method-level consensus rate across the seven score detectors is 84.8%, versus 62.7% for pressure and 57.7% for temperature. Nevertheless, high consensus rates for IMP and AAMP partly reflect their small, persistent event sets; a high rate should not be read independently of event count and raw Jaccard.

The symmetric consensus procedure yields 15 temperature, 41 total-precipitation, and 36 pressure intervals with at least three simultaneously active score detectors. Peak concurrence reaches five detectors for temperature and six for total precipitation and pressure. Examples of maximal concurrence are shown in Table 3; these dates are candidates for physical follow-up, not externally validated events.

3.3 Deep reconstruction diagnostics

Tables 4 and 5 report final standardized MSE and absolute errors after overlapping reconstructions are mapped back to the original variable scale. Reference-fit, validation, and evaluation losses are close enough to give no evidence of a large final-epoch generalization gap, although only one temporal split and one seed are available. Temperature TimesNet has the clearest validation increase (reference fit .00288, validation .00691), while its evaluation loss lies between them.

DLinear reconstructs all three series almost exactly, most strikingly for total precipitation. This is a property of the learned reconstruction mapping and the smooth/zero-heavy input, not evidence that DLinear has discovered a superior anomaly definition. Percentile thresholding still

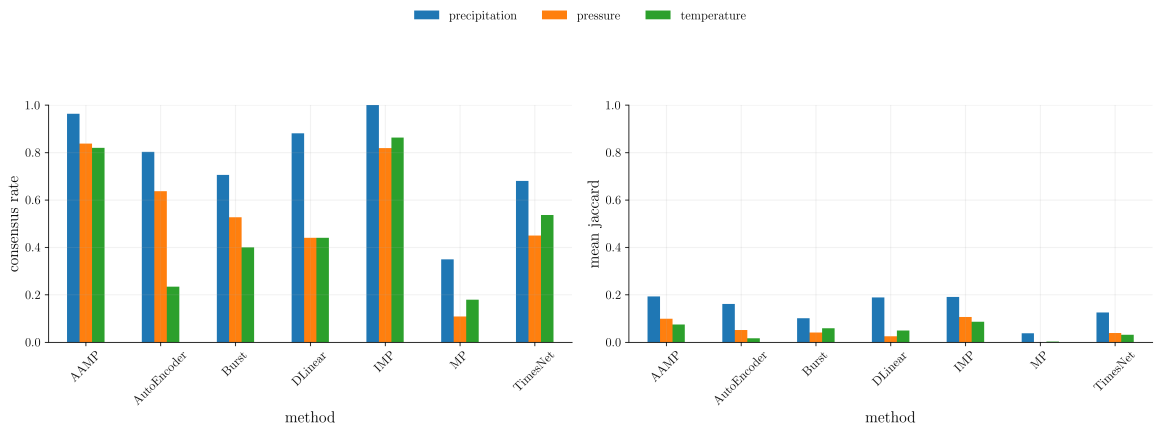


Figure 1: Cross-variable comparison of event count, consensus rate, raw Jaccard agreement, and score seasonality.

Table 3: Illustrative maximal-concurrence consensus peaks. The method list may include participation away from the peak within the merged interval.

Series	Peak date	Peak count	Participating methods
Temperature	2003-08-02	5	AAMP, DLinear, IMP, MP, iForest
	2019-07-08	5	AAMP, AE, DLinear, IMP, TimesNet, iForest
	2023-08-08	5	AAMP, DLinear, IMP, TimesNet, iForest
Total precipitation	2015-07-29	6	AAMP, AE, DLinear, IMP, TimesNet, iForest
	2015-12-26	6	AAMP, AE, DLinear, IMP, TimesNet, iForest
	2020-04-26	6	AAMP, AE, DLinear, IMP, TimesNet, iForest
Pressure	2000-04-01	6	AAMP, AE, DLinear, IMP, TimesNet, iForest
	2010-10-26	6	AAMP, AE, DLinear, IMP, MP, TimesNet, iForest
	2017-11-29	6	AAMP, AE, DLinear, IMP, TimesNet, iForest

ranks its very small residuals and selects 88 days. The selected dates can consequently be sensitive to numerical or subtle modelling differences. More generally, losses cannot be ranked across architectures as a substitute for anomaly quality: each architecture has a different reconstruction bias, and the physical-scale errors cannot be ranked across variables without common units and scientific tolerances.

4 Analysis by variable

4.1 Temperature

Temperature has the weakest score-detector consensus of the three variables. IMP, AAMP, and Isolation Forest form a modest amplitude/local-pattern group, whereas MP and the deep detectors contribute largely distinct raw days. The five DLinear events arise from strongly seasonal and temporally clustered residuals; the 67 AutoEncoder events are far more dispersed. The maximal multi-method peaks occur in boreal summer in the current record, but no heat event attribution should be made without the station/grid metadata and an independent catalogue. The event-month histogram and folded interpretations should be used to distinguish a seasonal opportunity effect from exceptional within-season values.

4.2 Total precipitation

Total precipitation produces both the most burst intervals (252) and the highest cross-method event consensus. Its raw Jaccard matrix separates an amplitude/local-window family (IMP,

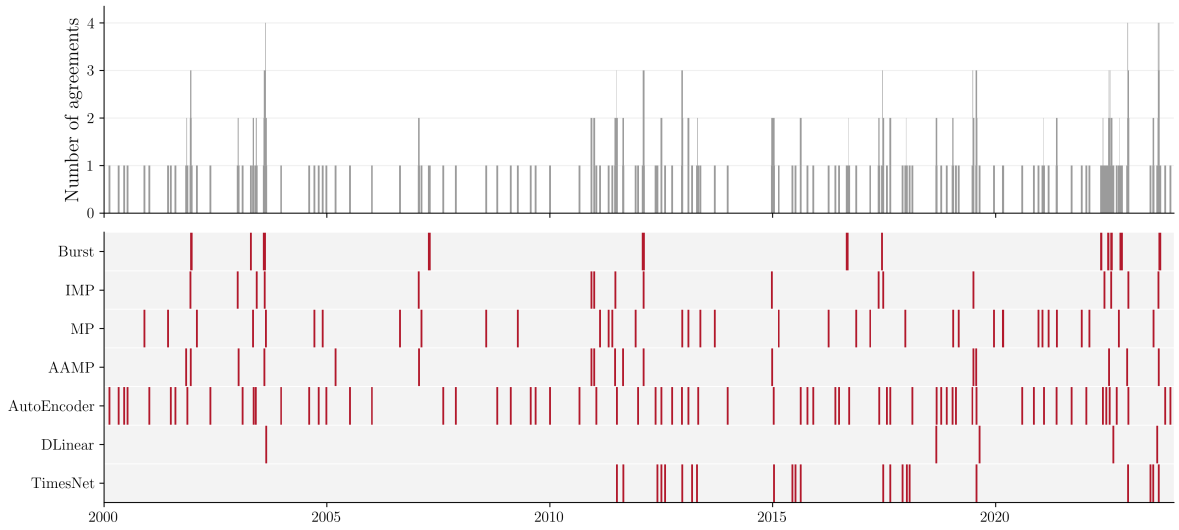


Figure 2: Temperature event timeline and daily method agreement.

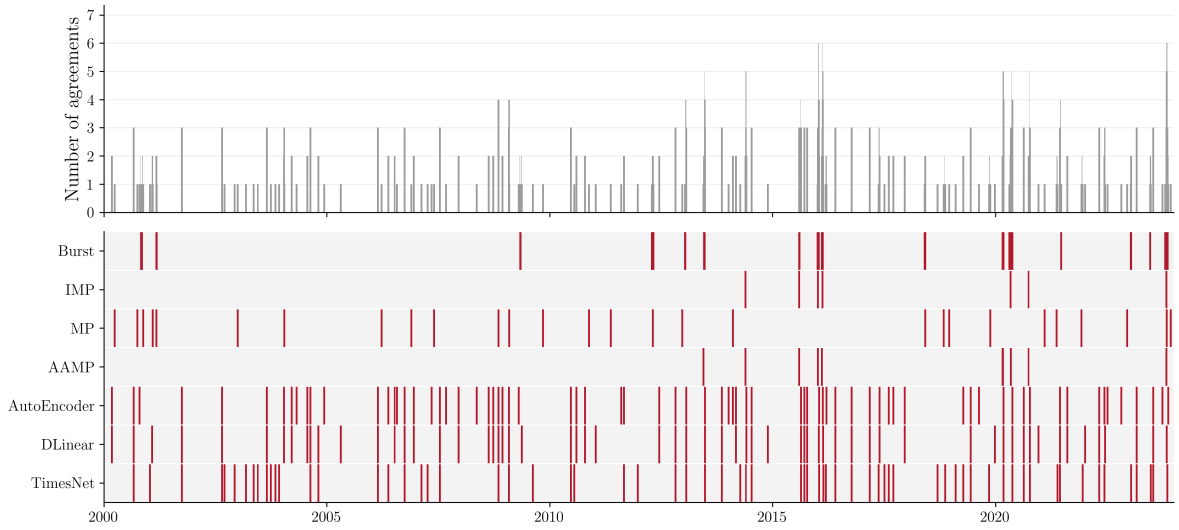


Figure 3: Total-precipitation event timeline and daily method agreement.

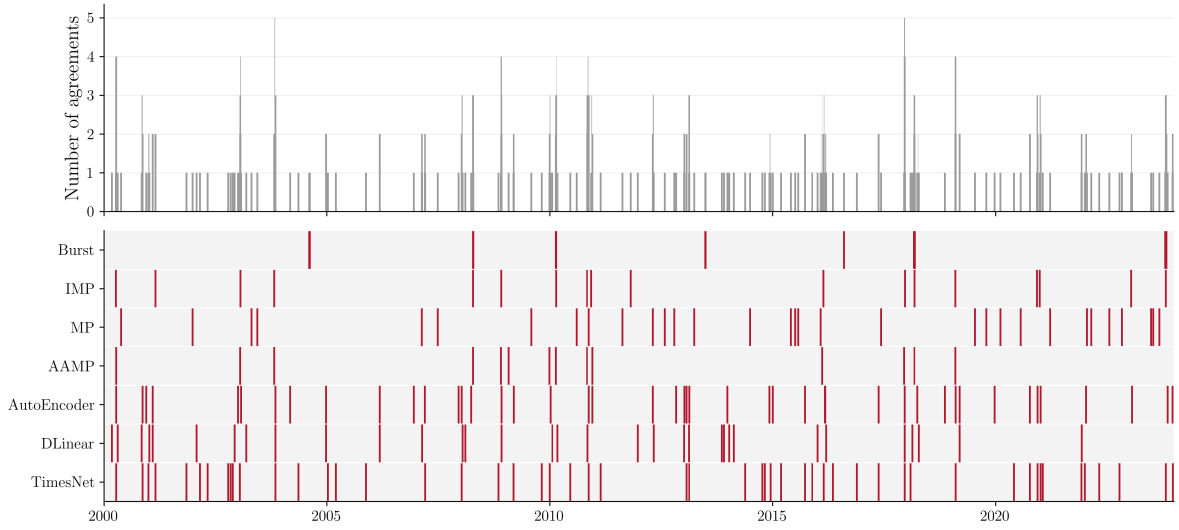


Figure 4: Pressure event timeline and daily method agreement.

Table 4: Final deep-model mean squared error on standardized data.

Series	Model	Reference-fit MSE	Validation MSE	Evaluation MSE
Temperature	AutoEncoder	.05936	.05698	.06204
	DLinear	2.24×10^{-4}	1.03×10^{-4}	1.19×10^{-4}
	TimesNet	.00288	.00691	.00429
Total precipitation	AutoEncoder	.53821	.52794	.52739
	DLinear	6.94×10^{-14}	6.64×10^{-14}	6.76×10^{-14}
	TimesNet	.00723	.00257	.00361
Pressure	AutoEncoder	.18981	.17094	.20311
	DLinear	1.19×10^{-6}	8.62×10^{-7}	9.10×10^{-7}
	TimesNet	.00854	.00449	.00475

Table 5: Absolute errors of the overlapping reconstruction time series. Values retain each variable's original scale.

Series	Model	Median $\lvert e\rvert$	Mean $\lvert e\rvert$	Maximum $\lvert e\rvert$
Temperature	AutoEncoder	.9151	1.1165	6.8143
	DLinear	.0030	.0033	.0276
	TimesNet	.0420	.0959	2.8480
Total precipitation	AutoEncoder	.00977	.02152	.42584
	DLinear	1.27×10^{-8}	1.87×10^{-8}	6.67×10^{-7}
	TimesNet	.000754	.001264	.077997
Pressure	AutoEncoder	201.17	252.06	1691.08
	DLinear	.3335	.4727	4.2362
	TimesNet	20.7288	28.6286	576.405

AAMP, Isolation Forest) from a reconstruction family (AutoEncoder, DLinear, TimesNet). This suggests that the same broad episodes can contain two different signals: concentrated local extremes and distributed reconstruction departures. MP remains more specific, with only 43.6% timeline consensus and low raw Jaccard. Because total precipitation is often zero-heavy and intermittent, the nearly exact DLinear reconstruction and the global percentile threshold require particular caution; inspection should report wet-day frequency and accumulation magnitude before calling these events meteorologically exceptional.

4.3 Pressure

Pressure has intermediate event consensus but the highest mean score seasonality. IMP, AAMP, Isolation Forest, and TimesNet each have F_S near or above .47, so their global 99th percentile emphasizes seasons in which their score distributions are naturally broader. AAMP has the highest timeline consensus (93.3%), while MP is almost entirely method-specific (11.9% and zero raw overlap with the other score detectors). The distinction is scientifically plausible as a distinction between level/amplitude anomalies and unusual normalized shape, but confirming it requires examining pressure tendency and synoptic context around the displayed subsequences.

5 Validity, sensitivity, and next analyses

The present run supports a comparative detector analysis, not a calibrated estimate of anomaly-detection accuracy. Its main limitations are:

- no labelled truth or independent event catalogue;
- one reference/evaluation boundary, one random seed, and final-epoch deep models;
- dependence among overlapping windows and among days within a cluster;

- fixed 15-day classical and 96-day deep contexts;
- a single global 99th-percentile rule despite variable score seasonality;
- no uncertainty intervals for event counts, Jaccard, or consensus; and
- missing source metadata, units, quality-control history, and spatial support in the report inputs.

The priority sensitivity analysis is a grid over percentile $(.975, .99, .995)$ and declustering distance (for example 7, 15, and 30 days), with year-block bootstrap intervals for Jaccard and consensus. Deep models should be repeated over multiple seeds and compared with a simple persistence or seasonal-climatology reconstruction baseline. External event validation criteria should be fixed before inspecting candidate dates. These additions would reveal whether the observed family structure is stable or is an artifact of one tail budget and one temporal scale.

6 Reproducibility

The numerical results used here are stored below `outputs/figures/` in one directory per variable. The principal machine-readable files are:

- `peak_diagnostics.csv`, `detector_jaccard.csv`, and `agreement_summary.csv`;
- `consensus_events.csv` and `deep_losses.csv`; and
- `deep_reconstruction_summary.csv`.

Cross-variable summaries and figures are in `cross_series/`. A complete archive should additionally record the repository revision, environment, hardware, input checksums, runtime, and full source metadata.

References

[1] F. T. Liu, K. M. Ting, and Z.-H. Zhou, Isolation Forest, *Proceedings of the IEEE International Conference on Data Mining*, 2008.

[2] C.-C. M. Yeh et al., Matrix Profile I: All Pairs Similarity Joins for Time Series, *Proceedings of the IEEE International Conference on Data Mining*, 2016.

[3] G. E. Hinton and R. R. Salakhutdinov, Reducing the Dimensionality of Data with Neural Networks, *Science*, 313(5786):504–507, 2006.

[4] A. Zeng et al., Are Transformers Effective for Time Series Forecasting?, *Proceedings of the AAAI Conference on Artificial Intelligence*, 2023.

[5] H. Wu et al., TimesNet: Temporal 2D-Variation Modeling for General Time Series Analysis, *International Conference on Learning Representations*, 2023.

[6] R. B. Cleveland, W. S. Cleveland, J. E. McRae, and I. Terpenning, STL: A Seasonal-Trend Decomposition Procedure Based on Loess, *Journal of Official Statistics*, 6(1):3–73, 1990.